ShaderPit

USER DOCUMENTATION

Procedural material effects, material layering and look development

MATERIAL LOOK DEVELOPMENT

DOCUMENT DETAILS	
Version	0.9.79
Blender compatibility	Blender 5.2 (declared in add-on metadata)
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Installation, feature reference, workflows and troubleshooting.

1. Installation

The add-on is distributed as a single Python file. Install it from Blender Preferences using the add-on installation workflow available in your Blender version.

- Open Blender 5.2.
- Go to Edit → Preferences.
- Open the add-on installation interface (Get Extensions / Install from Disk, depending on the Blender build).
- Select **shaderpit_v0_9_79_STABLE_AUTHOR_FIX.py**.
- Enable the add-on after installation if Blender does not enable it automatically.

Interface location: 3D Viewport → Sidebar (N) → ShaderPit

If the add-on does not appear, check the Blender version, confirm that the file is the actual .py add-on, and inspect the system console for registration errors.

2. Overview

ShaderPit is a material variation and look-development add-on for Blender. It combines object material management, procedural surface effects and a layered Material Mixer workflow in one panel. The documented build is version 0.9.79 and is marked as a stable release in the project history.

The panel is designed around an active mesh object and its active material. Select a mesh with a material to access the material effects and mixer controls.

3. Object Materials

The Object Materials section provides a Blender-style material workflow for the active mesh, including material selection and slot management. The source includes tools to create, rename, duplicate, delete, assign and manage material slots.

- **Create / Add Material:** create a material for the active object.
- **Rename:** change the selected material's name.
- **Duplicate:** create an independent copy and assign it to the active slot.
- **Delete:** remove the selected material datablock. Use care when deleting shared materials.
- **Save Material / Save as Asset:** save a material copy or write a material asset to the configured library folder.
- **Asset Browser:** open Blender's Asset Browser to browse saved material assets.

4. Effects

The Effects tab adds procedural effect groups to the active material's node tree. Effects can be enabled, disabled, reordered, collapsed, reset or removed using the controls in their headers.

Effect	Purpose / controls
Grunge	Grunge and alpha variation driven by texture/noise sources and adjustable mask controls. The add-on includes source-folder settings and random image/noise tools.
Dirt	Adds a dirt mask/effect group with adjustable mask controls.
Roughness	Adds roughness variation through a procedural mask.
AO	Adds an ambient-occlusion-driven effect/mask.

Effect	Purpose / controls
Color (HSV)	Adjusts hue, saturation and value as a material color effect.

Effect masks can be previewed using View Mask. Resource folder preferences allow ShaderPit to search configured Grunge/Alpha and texture folders, including subfolders.

5. Material Mixer

Material Mixer builds a layered material workflow. The active material is always the Base; additional layers are blended over it using their layer settings and masks.

- Add the Material Mixer from the Mixer tab.
- Add and reorder material layers; enable or disable individual layers.
- Adjust each layer's blend mode and amount.
- Use a layer mask to control where the layer contributes.
- Create a mask texture for a layer and use Live Paint to paint the mask on the object.
- Preview the active layer mask with View Mask when available.

Live Paint creates a 1024 × 1024 mask image and uses the object's UV map as the default mask mapping source. A suitable UV map is required.

6. Resource Folders and Mapping

Open the add-on preferences to configure the Grunge / Alpha and Textures source folders. ShaderPit searches their subfolders as well. Effects expose source, mapping, coordinate, tiling and mask controls where supported by the selected effect.

Use consistent folder organization and image names to make source selection easier. Generated or externally supplied image assets remain managed through Blender's image datablocks and the configured folders.

7. Suggested Workflow

- Select a mesh object and create or assign its base material.
- Open ShaderPit in the 3D Viewport Sidebar.
- Add the effects needed for the look (for example Grunge, Dirt, Roughness, AO or Color).
- Adjust each effect's source, mapping and mask settings; preview masks when needed.
- Open the Mixer tab to layer additional materials over the active material.
- Create a mask texture and use Live Paint when a hand-painted blend mask is required.
- Save the finished material or save it as an asset in the configured library.

8. Troubleshooting and Limitations

Issue	What to check
ShaderPit panel shows limited controls	Select a Mesh object and make sure it has an active material.
Effect cannot be added	Confirm an active material exists and that the same effect is not already present.

Issue	What to check
Source images are missing	Configure the Grunge / Alpha or Textures folder in add-on preferences and verify that images are present in those folders or subfolders.
Live Paint cannot start	The mesh needs a UV map. Create or unwrap UVs before creating/painting the mask texture.
Material asset cannot be saved	Choose a valid material library folder in the panel and verify that the folder is writable.
Mask preview is unavailable	Confirm the selected effect or mixer layer has a valid mask output.

ShaderPit · Pit Bibiloni · Version 0.9.79 · Blender 5.2 · Project status recorded as WIP during the documented development period.